Progressive Exploration of AI-Generated Image Analysis: Presenting Gender Differences and Stereotypes in Text-to-Image Models

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Abstract* - *Text-to-image models are a class of artificial intelligence systems that have the capability to generate visually appealing images based on textual descriptions. These models employ natural language descriptions as a reference to generate visual content, facilitating the functionality of diverse content-creation tools and applications. In essence, these systems serve to close the disparity between textual and visual art by the autonomous generation of images that correspond with the given language. This process introduces novel avenues for creative expression and automation within domains such as graphic design, entertainment, and related professions. The comprehension of the dissemination of these fabricated pictures, particularly with respect to potential gender stereotypes, continues to provide difficulties. The present study introduces an innovative approach, referred to as Presenting Gender Differences, which assesses gender representation through an extensive set of self-presentation characteristics. This investigation explores gender markers present in the input text, specifically referring to terms such as 'a woman' or 'a man'. In addition, we put forth a pair of novel measures, namely the Presenting Gender Differences (PGD) vector and PGD score, for the purpose of evaluating gender presentation disparities. Our proposed approach for assessing the fulfillment of three well-known text-to-image models demonstrates a stronger correlation with human evaluations compared to the current CLIP-based measures. Ultimately, we establish the capacity of our measures to extend to occupational preconceptions associated with gender and race.

Keywords: Text-to-image models, Presenting Gender Differences, Natural language processing

1. INTRODUCTION

Recent developments in AI had lead to the groundbreaking breakthrough of text-to-image model, which permit textual descriptions to be converted into images of exceptional quality. These models are widely used in many different content-creation tools and have a big impact on a lot of different downstream applications.

The distribution and depiction of gender traits in the images these models produce, however, is an essential aspect of these models that is still poorly understood, despite their potential to be transformational. In particular, it is imperative to look at the possibility that these outputs contain gender preconceptions. In light of this requirement, our initiative presents a novel paradigm known as "*Presenting Gender Differences (PGD)*". This paradigm adopts a fine-grained strategy, concentrating on aspects of self-presentation associated with gender identity in text-to-image models. Through a close examination of the gender depictors found in processed inputs, such as terms like "a woman" or "a man," we aim to measure the differences in frequency of presentation-related characteristics. These characteristics might apply to clothing items, such as "a shirt" as opposed to "a dress." We accomplish this by using human annotation as a methodological cornerstone, which enables us to efficiently capture these subtleties. In addition, our research presents a brand-new measure called PGD (Presenting Gender

Differences), which is intended to measure these distinctions quantitatively.

To enhance the scalability and efficiency of our investigation, we propose an automatic method for estimating these *Presenting Gender Differences*. Our automated PGD metric, developed through our approach, demonstrates superior connection with annotations made by humans compared to metrics according to present CLIP scores. Importantly, these findings hold uniformly throughout three cutting edge T2I models, underscoring the robustness and reliability of our approach.

In addition to elucidating regarding presenting gender differences within text-to-image models, our project extends its focus to show how our measurements may be used generally to the setting of gender stereotypes, particularly those associated with various occupations. By shedding light on these crucial aspects of AI-generated content, our work contributes significantly to advancing the understanding of bias and fairness in artificial intelligence. This, in turn, paves the way for more responsible and equitable AI development, making our findings highly relevant and impactful in the realm of contemporary AI research and application.

.1.1 Difficulties Faced in the Text-to-Image Models

Semantic Gap: Textual descriptions and images are fundamentally different modalities. Bridging the semantic gap between text & images is a significant challenge. Text may be highly abstract and open to interpretation, while images are concrete and contain vast amounts of visual information. Translating descriptive text into a visually accurate image can be complex.

Ambiguity and Subjectivity: Natural language is inherently ambiguous and subject to interpretation. Different people can have varying mental images when reading the same text. Text-to-image models must navigate this ambiguity and make decisions about what details to include in an image. Deciphering context and intent from text can be difficult, and different interpretations can lead to varying image outputs.

Data Quality and Quantity: Training text-to-image models requires large and high-quality datasets of paired textual descriptions and images. Acquiring such datasets can be challenging. The quality of the data is essential, as it can significantly impact the model's ability to generate accurate images. Moreover, collecting diverse and comprehensive data to cover a wide range of textual descriptions and visual concepts is time-consuming and resource-intensive.

1.2 Use of Text2image Models

Models of text2image have various practical applications across different domains. Few usual usage examples for text2image models are listed below:

1. **Material Creation:** Visual content may be routinely produced using text-to-image models. from textual descriptions. This is valuable in creative fields like graphic design, advertising, and content creation. For example, you can use these models to generate artwork, product images, or visual assets for marketing materials based on written descriptions.

2. **E-commerce:** Text-to-image models may be used to E-commercial items to generate product pictures based on product descriptions. This eliminates the need for costly picture sessions and allows firms to display their items online. Additionally, it can assist in tailoring product photos to satisfy certain client requirements.

3. **Videos and Online Games:** Video games and virtual worlds may be developed using text-to-image models. Systematically created materials are made possible by its attribute to create in-game objects, people, and possibilities according to narrative explanations. Thereby, game creation is more versatile and uses minimal resources compared to before.

4.**Cartoons and Narrative:** Text-to-picture models can be used by storytellers and pictorial creators to make illustrations for their works or

comic books. The genuine process may be spearheaded for artists by handing formulated descriptions of settings or people, which allows them to efficiently create graphic representations.

5.**Data Augmentation:** In machine learning and computer vision tasks, text-to-image models can be used to make greater datasets. For example, you can create non-natural images according to textual representations to create bigger, more expansive training datasets for picture classification or tasks involving finding objects.

6. **Design and Prototyping:** Architects, industrial designers, and engineers can use text-to-image softwares to picture and sample their thoughts. By depicting architectural or product designs in text, they can obtain visualizations, aiding with formulating and realizations,

7.**Availability and Incorporation:** Text-to-image models can help personnel with optical disabilities by transforming textual material into pictures or depictions. This can aid in creating content more accessible and inclusive.

8.**Education and Training:** In educational materials, text-to-image models can be used to create diagrams, charts, and educational illustrations based on textual descriptions. This aids in simplifying complex concepts and improving the comprehensibility of educational materials.

9. **Concept Realization:** Researchers and scientists can use text-to-image models to create visual depictions of formalized random ideas, hypothesis, or data. This can enable communication and comprehension within their respective areas of work.

Text-to-picture models have the prospect of streamlining content creation, enhancing transmission, and enabling multiple procedures to behave more efficiently and have accessible workflow across a large variety of uses. Their inventiveness and prospect to automate visual content creation make them a valuable tool in many industries.

1.3 Text-to-Image Models

An artificial intelligence (AI) system that creates optical pictures or illustrations from textual depictions are known as a text-to-image model. These models use DL techniques, particularly Conv-Neural-Network and GAN, to bridge the gap between natural language text and visual content.

1.3.1 Types of Text2Image models

Imagen: A paradigm developed by the research team from Google in association with the brain team.It depends on the effectiveness of diffusion models to generate high-fidelity images and expands upon the capabilities of big transformer language models for text comprehension.

DALL·E: A neural network developed by OpenAI It uses text captions to produce graphics for a variety of schemes that may be demonstrated in natural language. Using a dataset of text–image pairings, an enhanced GPT3 version was given the task of creating images from written descriptions.

CLIP: Also developed by OpenAI, CLIP is a versatile model that understands and connects text and images. It can be used for a variety of tasks, including generating images from text and text-based image classification.

Latent Diffusion Models: These are another type of Various models that are employed to create pictures from text.

StackGAN: A two-stage text-to-image generating approach is called StackGAN. Based on the written description, it first creates a low-resolution image then, in a second step, enhances it to a greater degree of detail, resulting in visuals that are more realistic.

BigGAN: While primarily known for generating images from variable noise vectors, BigGAN can also formulate pictures from inputted

descriptions by conditioning the generation process on textual inputs.

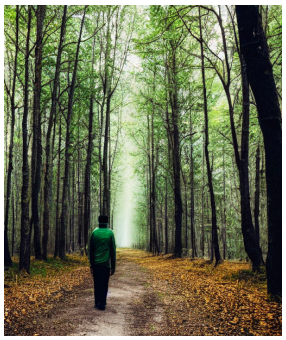
ProGAN(Progressive Generative Adversarial Network): Although primarily designed for image generation from noise, ProGAN can be adapted to formulate images from textual prompts by conditioning the model on text.

SAGAN(Self-Attention Generative Adversarial Network): SAGAN incorporates self-attentional processes within the GAN architecture, allowing it to generate high-quality images from text inputs with better coherence and attention to detail and more.

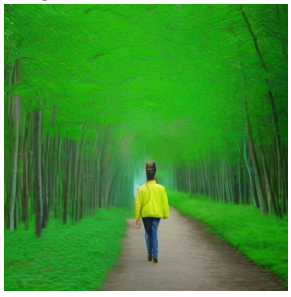
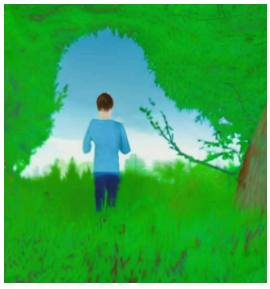
Example: Consider an example of the text:
“A man walking through a forest.”
The Image generated by some models are:
Dalle:




Stable Diffusion:

CogView2:

These are the images generated by some models based on gender stereotypes.

1.3.2 Machine Perception

The project “AI-Generated Image Analysis: Revealing Presenting Gender Differences and Stereotypes in Text-to-Image Models” is a fascinating intersection of Artificial Intelligence and the processing of natural languages (NLP). A subfield of artificial intelligence called "machine perception" enables machine entities the ability to analyze and comprehend data. from the real world in a manner akin to human perception. It circumscribes sub-fields like computer vision, which is pivotal for comprehension and formulating images, a vital aspect of this project.

Natural Language Processing (NLP), another sub-field of machine perception, plays a vital role in shedding light on the textual input that guides the image generation process. Advanced NLP techniques enable the model to understand context, semantics, and intricacies in the input text. In this project, we use these two areas of work to investigate how text-to-image models, which are an integral part of machine understanding, represent different genders based on documented descriptions. We provide a paradigm known as Presenting Gender Differences (GPD), which makes use of certain self-presentation attributes to delve deeper into these disparities.

We estimate the recurring variations in presentation focused elements (such as "a car" or "a dog") via human analysis by delving into the input text's gender signals (such as "a girl" versus "a boy"). We also present a brand-new metric: GPE (Gender Perception Equality), which measures the extent of gender bias in these models. This project not only contributes to our understanding of gender representation in AI-generated images but also offers insightful information about the relationship between NLP and machine perception. It emphasizes the significance of considering societal factors like

gender representation when designing and training AI models.

2. EXISTING MODEL

[01] **Analyzing Gender and Intersectionality in Machine Learning**: This system discusses how algorithms take into account social characteristics like sex, gender, and race. It highlights how important it is for teams to collaborate in order to clarify the goals of the system under development or study and assess any possible societal repercussions.

[02] **Analyzing Gender Representation in Multilingual Models**: This system studies the internal conceptions that provides transfer learning across different languages.

[03] CogView2: Using Hierarchical Transformers to Generate Text-to-Images Faster and Better : The CogView2 is a transformer with 6B:9B:9B parameters, designed for text2image generation in the general domain. This model addresses the slow generation and complexity issues associated with high-resolution images. Compared to its predecessor, Cog View, CogView2 can generate images with comparable resolution and quality at speeds up to 10 times faster when fine-tuned for fast super-resolution. It also naturally facilitates text-guided interactive picture manipulation.

[04] Text-to-Picture Model Alignment with Human Input: This project has showcased that calibrated with personage comments can vastly increase the image2text arrangement in 3 areas of work: producing materials with a stated count, color, or background. We examined various delineation choices (such as using an auxiliary loss and gathering diverse training data) and concluded that it is highly demanding to impartially gather the alignment-fidelity trade-offs without cautious examinations on such design alternatives.

[05] **The Elaboration Paradox: Text2Image Generation - Preethi, Singh, Elazar:** Text-to-image models can amplify gender biases in training data, but this can be reduced by accounting for discrepancies between training captions and model prompts.The methodology used in this are Stable Diffusion model , LAION-5B

[06] **MiZhang, Chanyel and Zipping Zhou(2021). Text2image production using multi generator text conditioned GAN:** In this project the researcher has used T2I model production with help of multi-generator.

[07] **Friendly Face: Do Text2Image Systems depend on Stereotype? - Khaln C. Fraser, Svetlna Kiritchnko, Iasr Negajholi:** Text-to-image systems can generate images that reflect social biases, such as stereotypes about different demographic groups. However, the Different models show uneven patterns.

[08] **Effortlessly Available Text2Image Amplifies Generation Demographic Stereotypes at Huge Scale - Fedricco Bianchi, Prattusha Kaluri:** Text-to-image models can amplify dangerous and complex stereotypes, even when the prompts are neutral or attempt to counter stereotypes.

[09] **Muse: Text2Image Generation via Generative Hidden Transformers - Huaiwen Chang, Han Zahng(2023):** Muse is a text-to-image Transformer model that is far highly well regulated and coherent than diffusion or autoregressive models, yet still delivers state-of-the-art picture generating capability. Fine-grained language comprehension and high-fidelity picture production are made possible by the deployment of a pre-trained LLM, which is drilled on a hidden modeling task in discontinuous token space.

[10] **Assorted language & learning of vision of general fashions concept:- Patrick Jhon, Guisepe:** Fashion CLIP is a Model groomed for the fashion business that resembles CLIP, using contrastive learning to generate transferable

representations of products. It can be used for retrieval, classification, and grounding.

Technique	Advantage	Disadvantage
DALLE-v2 , Stable Diffusion	Improved Training	Prompting Mitigation Limitations
CogView2	Speed and Quality	•Expensive High-Resolution Training

Table 1 : Advantages and Disadvantages of some popular techniques used in the Text2Image Models

3. PROPOSED SYSTEM

The proposed module in this project is called Presenting Gender Differences. This method analyzes gender representation in text-to-image models using comprehensive self-presentation features. Here's how it works:

Examine Gender Markers: The method examines gender markers within the text entry, such 'a woman' or 'a man'.

Measure Differences: It measures the differences in recurrence of human-centric qualities with human commenting, such as "a trouser" or "a watch."

Propose New Metrics: The method proposes two new metrics to assess Presenting Gender Differences: the PGD vector and PGDscore.

Automated Assessment: The method provides an automated way of assessing these metrics, which has been found to have a greater association with human annotations than existing CLIP-based measurements.

Generalization: The Metrics may be used to racial and gender stereotypes in the workplace.

This method opens up new possibilities for understanding the spread of images created by text-to-image models, especially in regard to potential gender stereotypes.

3.1 System Architecture

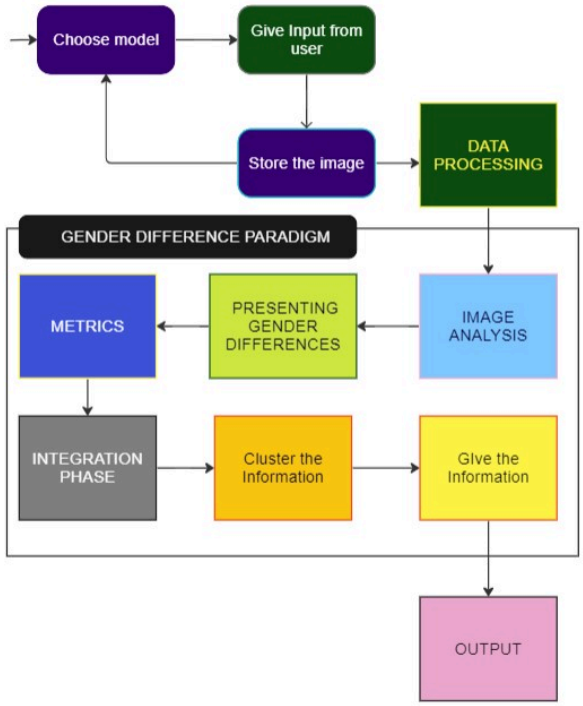


Fig 01 : Architecture of the proposed system

The project architecture shows the workflow of the project where initially we will give choose a model and give any text as input then that input is sent to the text encoder to map with the image encoder and the image is generated, further that image is stored and then the image is processed

to find the metrics of PGDto check the way where each model reinforces to learn on the stereotype defined.

The above architecture depicts various modules that is present in this project namely:

1.Data Processing Module

2. Image Analysis Module

3. PGDModule

4. Integration phase Module

4. RESULTS AND ANALYSIS

A module named Data Processing module was implemented and using which images were generated for initial study. The prompt was "A man riding a horse" and following are the different generated images by each model

Stable Diffusion:



Fig 02 : Images generated by Stable Diffusion

DALL-E:



Fig 03 : Images generated by DALL-E

CogView2:

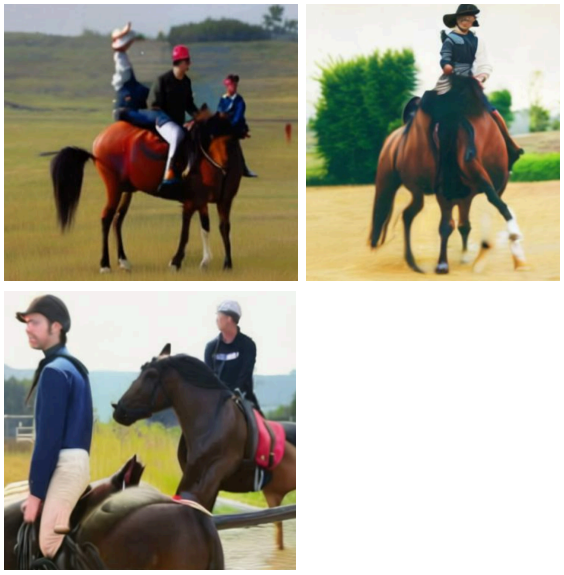
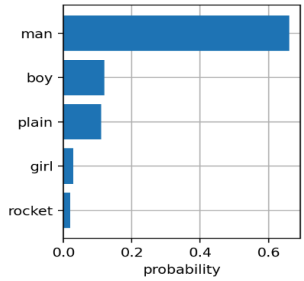


Fig 04 : Images generated by CogView2

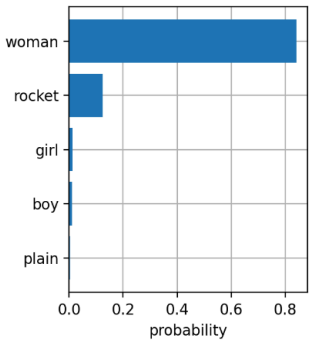
CLIP BASED OUTPUT

Considering an image of a portrait of an astronaut in front of the American flag and upon Zero Shot Classification:









Zeroshot classification is a paradigm that classifies images without any prior training or knowledge on the matter at hand.

5. CONCLUSIONS

In this initial phase, we've discussed the importance of text to image model and the potential issues related to gender stereotypes in the images they generate. The Presenting Gender Differences (PGD) metric was introduced as a tool to study and measure these biases, particularly in terms of presentation-centric attributes and the images for the analysis was derived from the data processing module and the CLIP module was also successfully implemented for the future analysis of CLIP score comparison.

6. FUTURE WORK

In Phase 1, the adversities encountered in the T2I were explained & their necessities for an improved system according to the gender disparities were put forth. Many published works and journals from the internet were referred to and goals were put forth. The architecture of the system and workflow were produced. Implementations of the work put forward and the analysis of the final result would be taken forward in the next phase of the project. Proceeding onto the second phase, the methods that we employed are spearheaded by a thorough work plan that specifies the mission, timeframes, and goals that will leverage our project towards success. The approach taken by us is supported by the research and findings presented in different published works and studies, offering a strong substructure for our project's procedure.

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